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https://github.com/Vinayak-YB/dllab-cv3

PHYSICS LEARNING IN VIDEO MODELS

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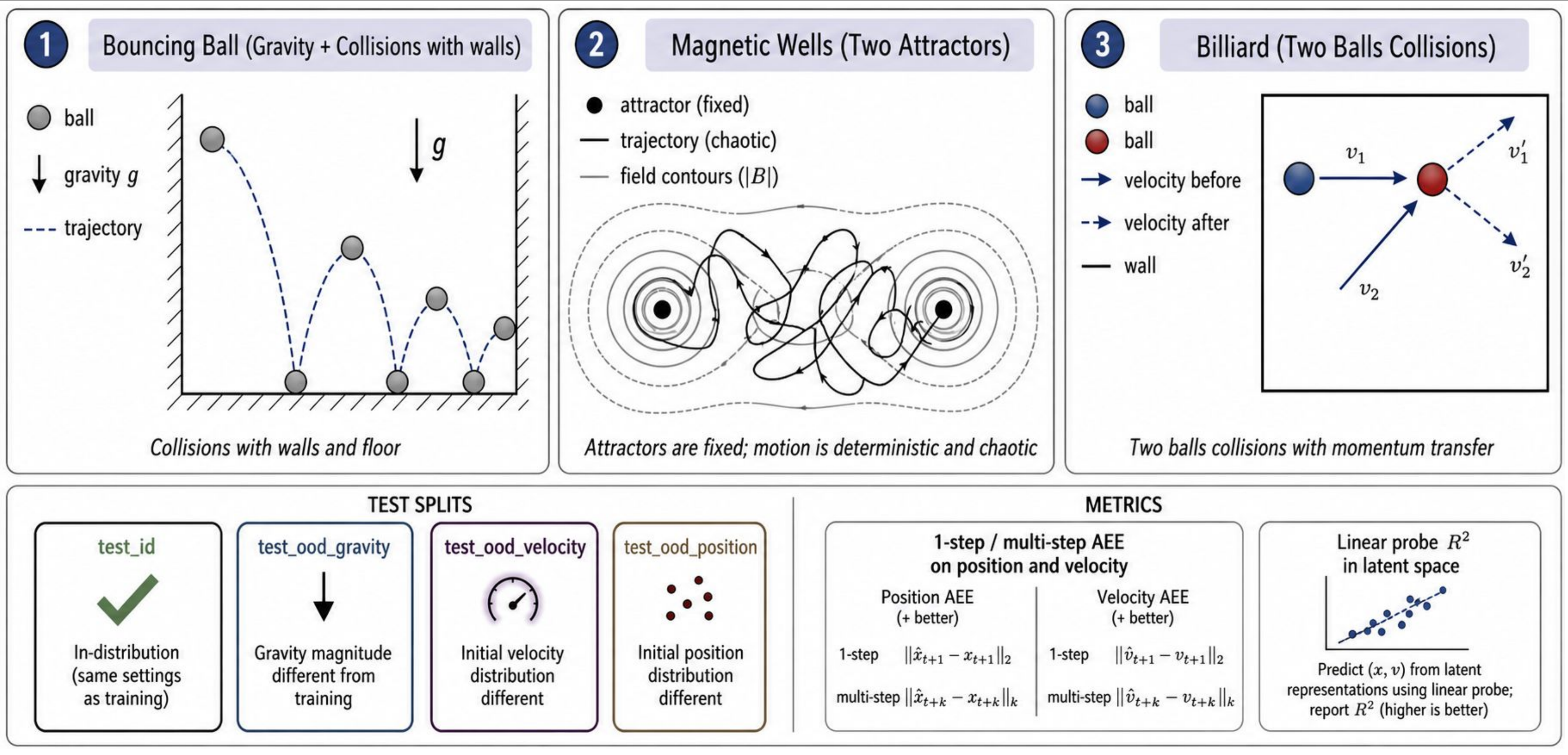
Introduction

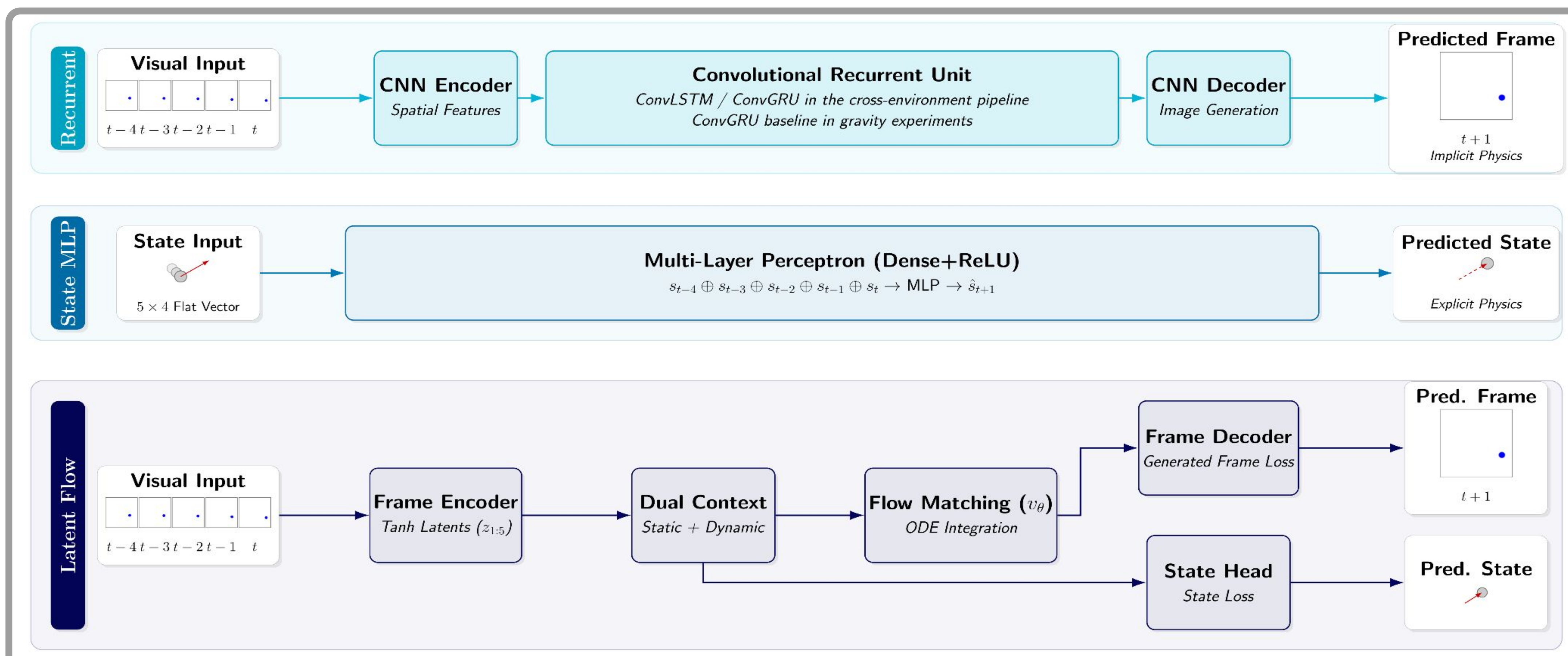
Problem: Do generative video models truly grasp physical laws (e.g., gravity, momentum) or merely replicate visual patterns?

Core Research Questions:

- How do generative architectures (Latent Flow Matching) compare to recurrent baselines (LSTM/GRU) at learning physical dynamics?
- Does an explicit state-based model (trained on ground-truth coordinates) generalize better to unseen scenarios than implicit visual/latent models?
- How robust are these models under out-of-distribution physical shifts (e.g., altered gravity, unseen velocities, or spawn distributions)?



Method



Experimental Protocol

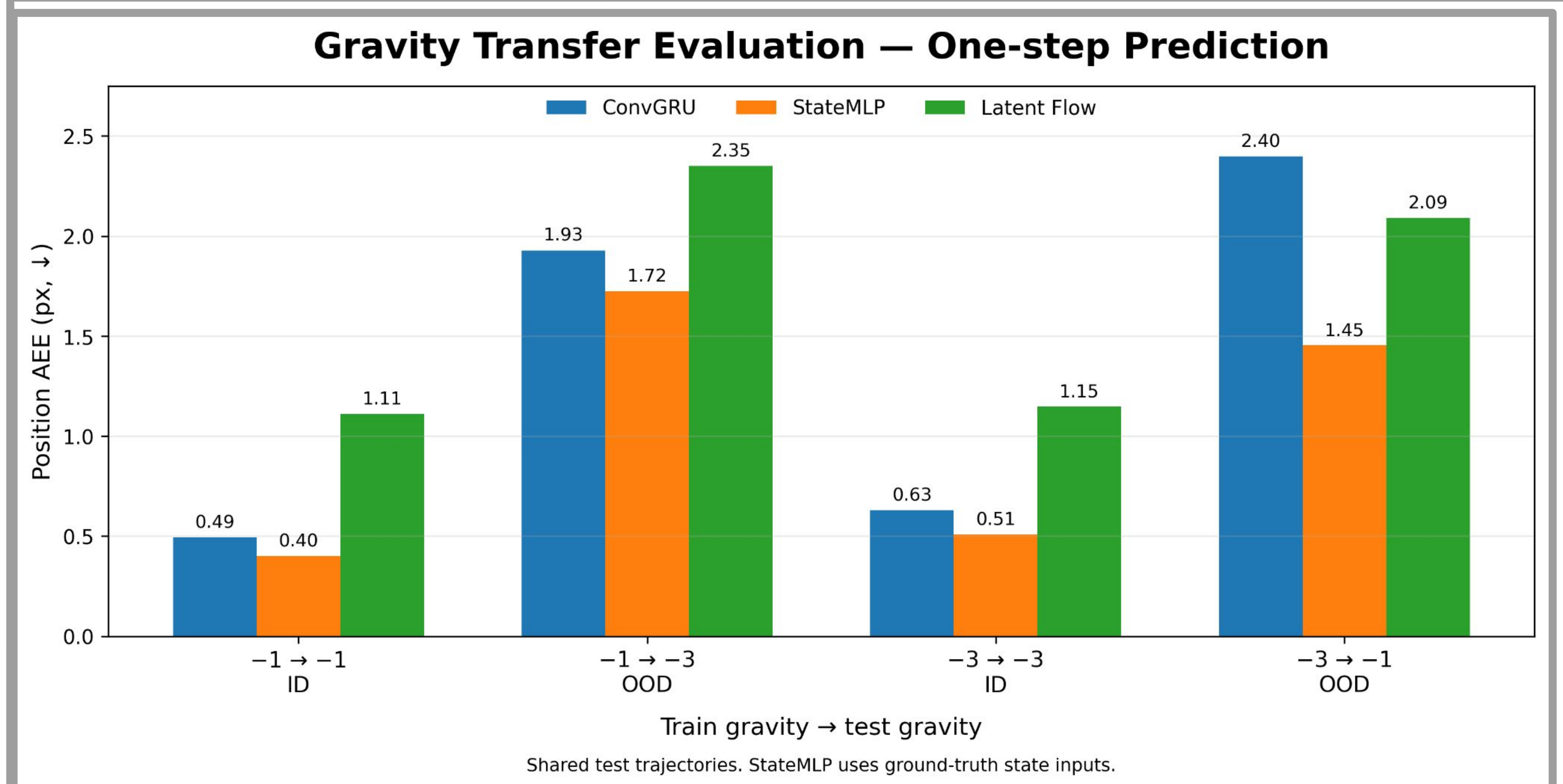
Input: 5-step frame or state context
Prediction: next frame and/or state
Evaluation: one-step and 10-step autoregressive rollout
Metric: observed-space Position AEE
Data: shared held-out test trajectories

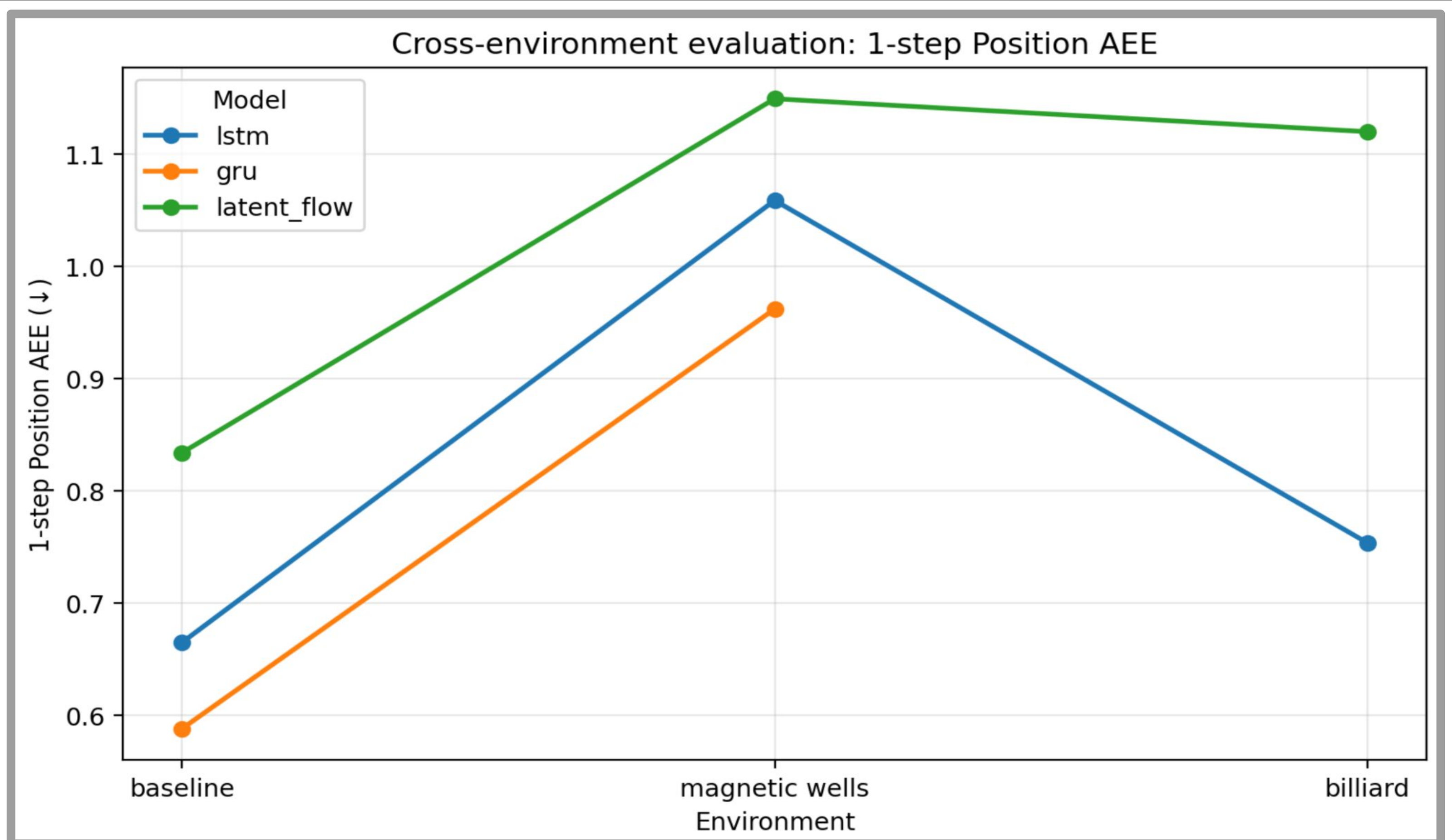
Evaluation suites Cross-environment: bouncing ball, magnetic wells, billiard

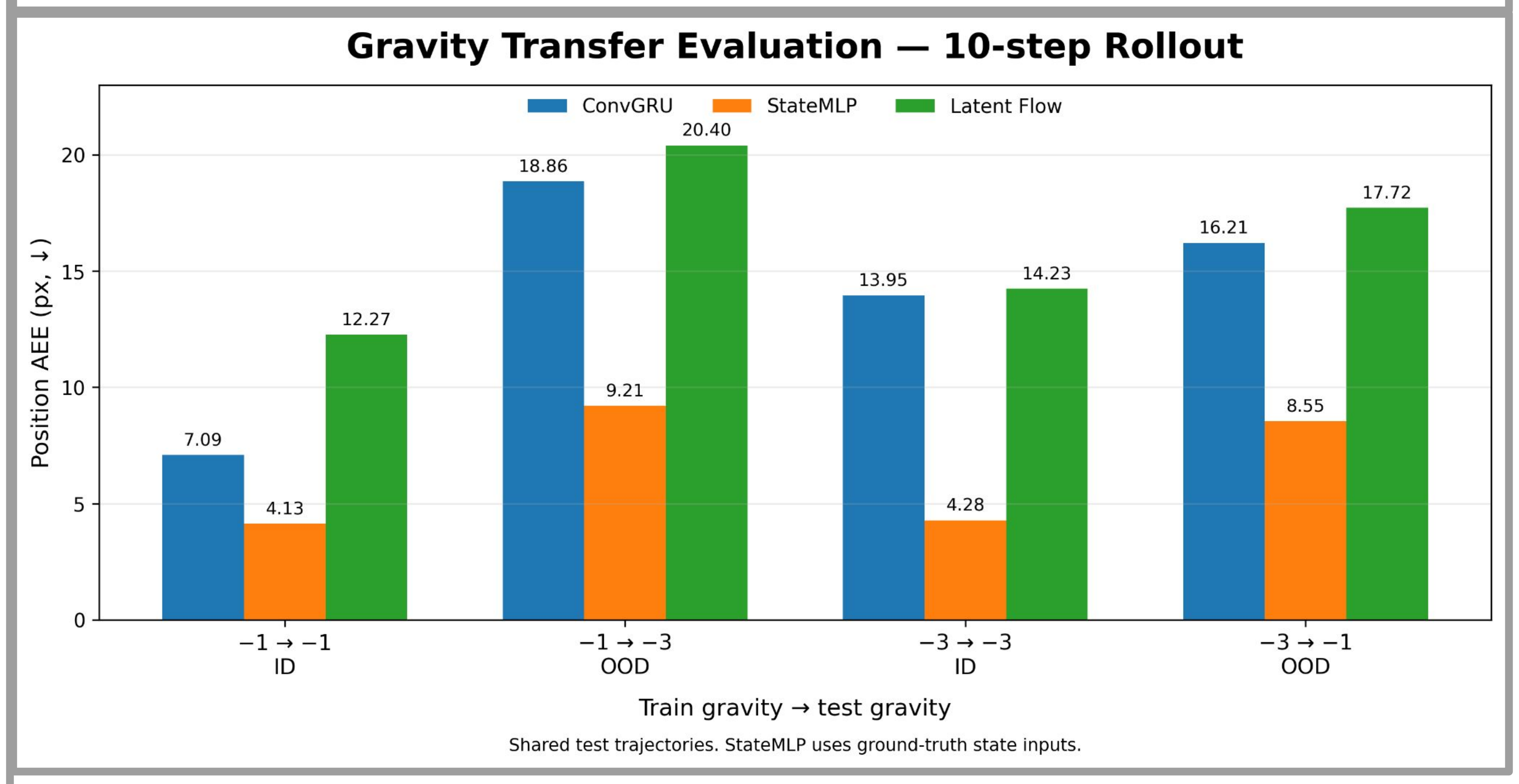
Gravity transfer: g=−1 ↔ −3

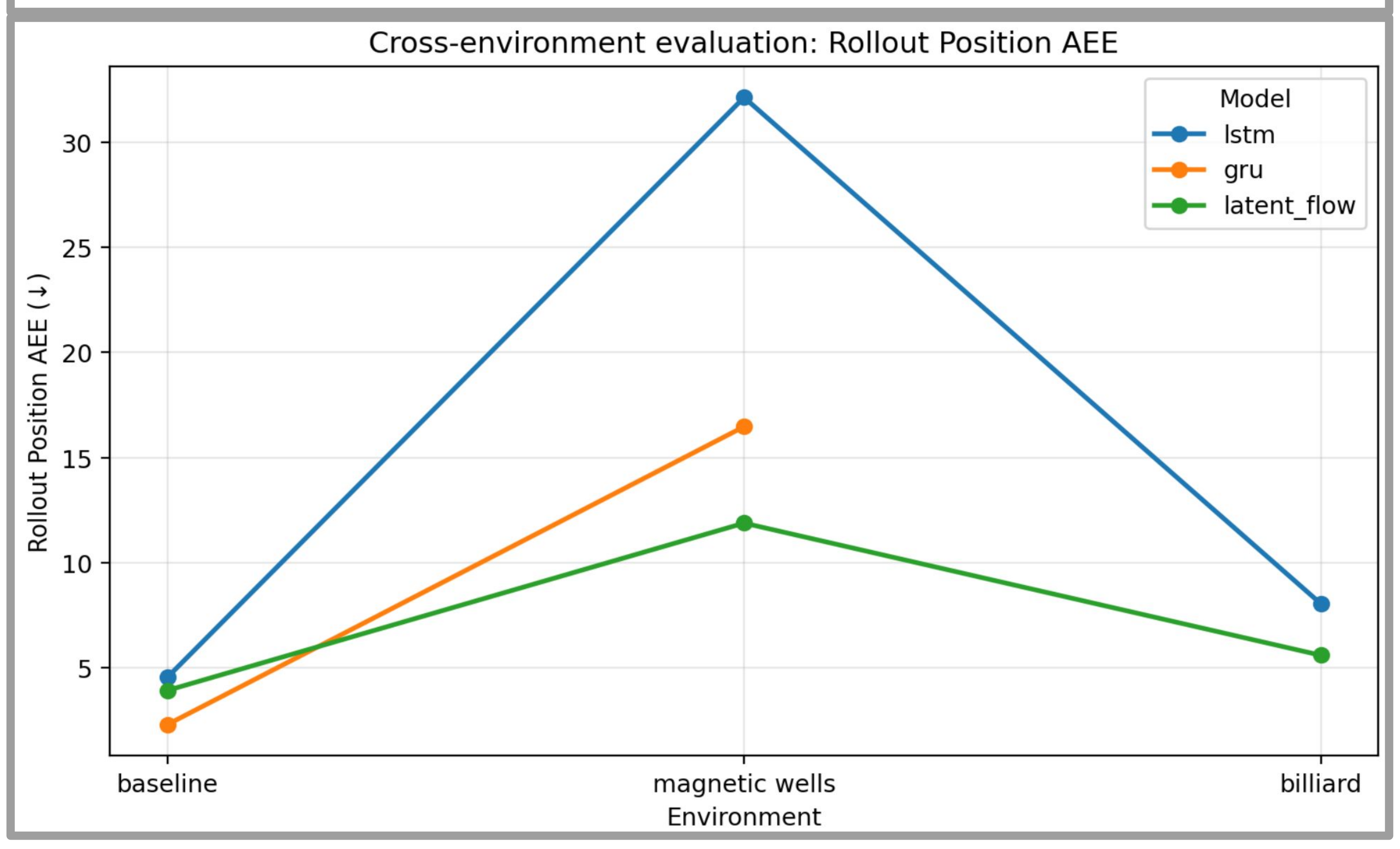
Latent Flow uses auxiliary state supervision during training.

Quantitative Results









[1] Physics learning in Bouncing Ball environment: https://github.com/eliaskempf/world-model-physics
[2] Orbis Flow Matching Video model: https://github.com/lmb-freiburg/orbis
[3] ByteDance paper studying studying physics learning in video models: https://arxiv.org/pdf/2411.02385